

Dr Amar Banerjee

Bridging research curiosity, inventive thinking and real-world impact



■ My story of innovation

My journey began in control and automation research, where I learned that a technically elegant answer is valuable only when it addresses a problem that truly matters. Across industrial systems, healthcare, AI and enterprise innovation, one question remained constant: **how does an idea travel from insight to experiment, patent, partnership and impact?**

The biggest barrier is rarely a shortage of ideas. It is often the absence of a shared language between researchers, students, institutional leaders and industry teams. These programs are designed to create that language together.

■ The experience I promise

1 • BEGIN WITH CURIOSITY

Start with lived experiences, tensions, anomalies and unmet needs - not a pre-selected solution.

3 • INVENT WITH DISCIPLINE

TRIZ, reverse brainstorming and structured experimentation turn imagination into defensible concepts.

2 • MAKE PROBLEM VISIBLE

Stories, evidence and stakeholder perspectives reveal what conventional framing misses.

4 • LEAVE WITH NEXT MOVE

Every participant converts learning into a tangible canvas, prototype, pitch or action roadmap.

■ A shared innovation vocabulary

Academia often begins with knowledge; industry often begins with value. A shared sequence makes collaboration easier to initiate and sustain. **PROBLEM** worth solving **EVIDENCE** that matters

TENSION to resolve **INVENTION** with novelty

EXPERIMENT to learn **IMPACT** to scale

My facilitation promise

The room will be safe enough to question assumptions, rigorous enough to challenge easy answers and practical enough to produce something participants can use the next morning.

How I facilitate innovation

Research depth, invention experience and a practical path from problem to proof.

■ What I bring into the room

- 12+ years across research, applied AI and industrial R&D
- 30+ publications and 25+ filed patents
- 7 granted patents
- Experience connecting technology, research rigor and human-centred innovation
- Teaching, facilitation and mentoring across academic and professional cohorts

■ My promise to students

You will not be asked merely to “think outside the box.”
You will learn how to find a better box: a meaningful problem, a useful contradiction and a credible path from idea to opportunity.

■ How we will work together

- Short stories and cases that make complex ideas relatable
- Individual reflection followed by collaborative studio work
- Frameworks used on participants’ own academic and professional challenges
- Constructive critique that improves novelty, relevance and clarity
- Reusable canvases that continue after the workshop

■ My invitation to academic leaders

Let us create a campus culture where publication is not the finish line, patents are not isolated trophies and industry engagement is not transactional. Together, we can help students and faculty connect knowledge, invention and societal value through a vocabulary everyone can use.

CREDIBILITY BEHIND THE PRACTICE

PhD, IIT Hyderabad • IEEE Senior Member
Visiting Professor, COEP Technological University
Reviewer for international journals and conferences

■ The shift I want to enable

Publication as the endpoint	→	Knowledge as a starting point
Research topics inherited	→	Problems actively discovered
Ideas discussed and admired	→	Experiments designed and tested
Industry invited at the end	→	Partners engaged around shared value

START WITH ONE COHORT

Choose one capability. Define one visible change. Run the pilot. Review the evidence.

Three pillars of innovation

An idea becomes strategically interesting when it is **unique**, **hard to do** and **value adding** at the same time. The programs build student capability across all three.

01 UNIQUENESS

Meaningfully different

Students learn to ask whether the problem, insight or solution is genuinely different from what already exists - not merely a new implementation of a familiar idea.

SKILLS BUILT

Problem discovery, reframing, prior-art awareness, assumption breaking, divergent thinking and TRIZ.

02 HARD TO DO

Technically defensible

Innovation needs technical or execution depth. Students learn to work with constraints, contradictions, evidence and experiments so the idea is more than a superficial concept. **SKILLS**

BUILT

Systems thinking, technical trade-offs, experimentation, prototyping, non-obviousness and execution discipline.

03 VALUE ADDING

Creates real value

A clever idea is not automatically valuable. Students learn to connect innovation to a real user, institution, industry or societal need and define the change it should create. **SKILLS**

BUILT

Stakeholder insight, outcome framing, evidence, adoption thinking, value articulation and storytelling.

■ Why all three have to come together

Unique + valuable, but easy to copy

Interesting, but difficult to defend or sustain.

Unique + hard, but low value

Technically impressive, but unlikely to matter outside the project.

Hard + valuable, but not unique

Useful engineering, but limited differentiation or innovation claim.

Unique + hard + value adding

A stronger foundation for defensible innovation and meaningful impact.

■ How I help students build the three muscles

DISCOVER

Find tensions, unmet needs and consequential problems.

CHALLENGE

Question assumptions and seek a genuinely different angle.

BUILD

Work through constraints, contradictions and technical depth.

VALIDATE

Test evidence, stakeholder value, utility and feasibility.

COMMUNICATE

Turn the work into a research, patent or industry story.

THE CORE HABIT

Do not ask only, "Can we build it?" Ask: "Is it different? Is it difficult enough to be defensible? Does it create enough value to matter?"

What the three pillars are designed to enable

The objective is not idea generation for its own sake. Students learn to convert differentiated, substantive and useful ideas into stronger research, invention and industry-facing evidence.

■ Outcome 01 - Impactful research

UNIQUENESS

A consequential question, clearer research gap and a contribution that is meaningfully different.

HARD TO DO

Technical depth, rigorous methods, experiments, difficult trade-offs and credible evidence.

VALUE ADDING

A problem that matters to a field, stakeholder, industry, society or future system.

STUDENT-OWNED EVIDENCE

Problem landscape • novelty map • research question • experiment plan • contribution statement • publication pathway.

■ Outcome 02 - Patents and defensible IP

UNIQUENESS

Prior-art awareness and a clearly differentiated mechanism, method or system element.

HARD TO DO

Non-obvious technical substance, constraints and implementation depth that make the idea defensible.

VALUE ADDING

Utility, adoption potential and a reason the invention should exist beyond novelty alone.

STUDENT-OWNED EVIDENCE

Invention canvas • prior-art search terms • novelty hypothesis • non-obviousness rationale • prototype or evidence plan.

■ Outcome 03 - Industry-ready innovators

UNIQUENESS

Spot opportunities others miss and frame a differentiated response rather than repeating obvious solutions.

HARD TO DO

Move from idea to execution: handle constraints, collaborate, experiment, learn and improve.

VALUE ADDING

Connect technical work to business, user and societal outcomes and explain why it matters.

STUDENT-OWNED EVIDENCE

Problem brief • prototype or concept • decision rationale • outcome story • pitch • 30/60-day action plan.

WHAT ACADEMIC LEADERS CAN REVIEW AFTER A COHORT

Problem portfolio • research directions • possible IP candidates • experiment plans • industry-facing pitches • participant capability evidence • recommended next steps.

The program builds capability and stronger pathways. Publication, patent and hiring outcomes are not guaranteed and depend on the quality of subsequent work, evidence, review and execution.

LET'S BUILD YOUR INSTITUTION'S NEXT INNOVATION PIPELINE

From a one-day workshop to a repeatable campus capability



■ The collaboration pitch

Institutions rarely lack talent, ideas or ambition. The missing link is often a shared, repeatable pathway that connects academic curiosity to a problem worth solving - and then carries that problem towards an experiment, intellectual property, funding, an industry partnership or measurable student opportunity.

I collaborate with academic leaders to design learning engagements around the institution's real priorities. The aim is not to conduct another isolated event. It is to leave behind a common vocabulary, practical tools, visible outputs and the confidence to keep innovating.

■ Value-led engagement options

- 01 Signature Workshop | One day:** Create a shared vocabulary, practise the frameworks and leave with participant-owned canvases, concepts and action plans.
- 02 Capability Sprint | Modular series:** Combine workshops, guided application and a review clinic so faculty or student teams can apply the methods to live institutional challenges.
- 03 Innovation Partnership | Custom pathway:** Integrate challenge labs, research-to-IP clinics, leadership conversations and follow-through support around a defined campus or industry priority.

Formats can include faculty development, student bootcamps, research clinics, industry-academia challenge labs and leadership roundtables.

■ A simple path from conversation to impact

01 LISTEN

Clarify the audience, institutional priority and desired outcome.

03 FACILITATE

Create an energetic studio where participants build together.

02 DESIGN

Tailor cases, tools, challenges and participant deliverables.

04 SUSTAIN

Provide canvases and a follow-through roadmap for continued action.

What the institution can gain

A collaboration is designed around visible capability, reusable tools and a practical path for continuation after the event.

■ What the institution can gain

- A visible portfolio of novel, relevant problems
- Stronger patent, proposal and funding concepts
- Better research stories for industry and government stakeholders
- Employer-relevant evidence of student capability
- Interdisciplinary teams aligned around shared challenges
- Reusable innovation tools that extend beyond the event

■ Value-based scoping

Custom programs can be discussed based on the cohort, institutional priorities and desired outcomes. Each engagement is shaped around outcomes, customisation and follow-through - not delivery hours alone.

■ Program portfolio

01 • INNOVATION IN ACADEMIA

Find consequential problems; move from incremental publication to patent, partnership and funding pathways.

02 • THE MODERN-DAY PROFESSIONAL

Build judgement, narrative and inventive confidence needed to convert knowledge into opportunity.

03 • CONNECTING TECHNOLOGY TO EMOTIONS

Create human-centred technology concepts that people can trust, value and adopt.

04 • THE ART OF STORYTELLING IN EDUCATION

Turn ideas, projects and research into narratives people can understand, remember and act on.

Start the conversation

Share your priority, participant profile and intended outcome. I will help shape an engagement relevant to your campus and designed around tangible next steps.

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LOW-RISK STARTING PATH

Choose one cohort. Define one measurable capability. Run the pilot. Review the evidence. Scale what works.